Grading Rubric - DSA Final Assignment

Question 1: Dijkstra's Algorithm (25 Points)

- (5 pts) Clear explanation of the algorithm's greedy approach and relaxation.
- (5 pts) Correct explanation of why negative weights break the greedy assumption.
- (5 pts) Accurate step-by-step trace of the algorithm on the provided graph, showing correct intermediate distances.
- (5 pts) Syntactically correct and logical Python implementation using a priority queue.
- (5 pts) Accurate time complexity analysis (O(E log V) or O(V^2)) and space complexity (O(V)).

Question 2: Knapsack Problem (25 Points)

- (5 pts) Accurate mathematical formulation (maximizing sum of values subject to weight constraint).
- (10 pts) Correct recurrence relation with well-defined base cases and recursive step.
- (10 pts) Correct construction of the DP table for the specific example and the correct final maximum value (220).

Question 3: AVL Trees (25 Points)

- (5 pts) Correct definition: height difference of left and right subtrees is at most 1.
- (10 pts) Clear explanation and examples of LL, RR, LR, and RL rotations.
- (10 pts) Correct final tree structure after all insertions and accurate identification of the rotations performed (e.g., RR rotation after inserting 30, RL rotation after inserting 25).

Question 4: System Design and Concurrency (25 Points)

- (5 pts) Clear explanation of deadlock.
- (5 pts) Accurate listing and explanation of the 4 Coffman conditions (Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait).
- (5 pts) Clear explanation of the Bakery Algorithm ticket numbering system.
- (10 pts) Correct implementation of a thread-safe Singleton and a solid explanation of double-checked locking and its pitfalls (e.g., volatile keyword in Java).